

ROSHAN RAJ MAHATO

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EDUCATION

Ramgarh Engineering college
Computer Science & Engineering

Jun 2021-Jul 2025
7 CGPA

SKILLS

Programming: Python, C++, **Object-Oriented Programming**, **Data Structures**

Machine Learning: **Scikit-learn**, Logistic regression, SVM, Random Forest, XGBoost & other algorithms

Deep Learning: **TensorFlow**, PyTorch, CNNs, RNNs, **Transformers**

AI/MLOps: **MLflow**, Docker, AWS (EC2, ECR), GCP (Cloud Functions, Cloud Tasks, Buckets) Git, **CI/CD**

Generative AI: **LangChain**, RAG, LLMs, OpenAI

Data Science: NumPy, Pandas, Matplotlib, **Statistical Analysis**

INTERNSHIP

Medical Chatbot for Context-Aware Medical Information Retrieval | ybifoundation(remote) jun-jul 2024

- Developed a Medical Chatbot leveraging the **Llama-2-7B-Chat** model for accurate, context-aware medical information retrieval.
- Utilized LangChain Framework to streamline generative AI workflows, enabling seamless integration of **the Llama-2 model and Pinecone database** for efficient medical query handling.
- Integrated Pinecone Vector Database for **efficient storage and retrieval of embeddings**, enhancing response time and accuracy.
- Built a user-friendly web interface using **Flask, HTML, and CSS** with a **modular backend** for maintainable and scalable code structure.

PROJECTS

Student Performance Prediction

Dec 2023

- Achieved a 96% accuracy rate in forecasting student academic performance by developing and deploying a machine learning model.
- Managed data integrity by handling missing values and encoding categorical variables, enhancing quality by 33%.
- Conducted experiments with both classification and regression algorithm to identify the most suitable approach.

Credit Card Fraud Detection

Oct 2023

- Developed and fine-tuned a **logistic regression**-based machine learning model achieving an 87% accuracy rate.
- Analyzed transaction patterns to identify anomalies and distinguish between genuine and fraudulent activities.
- Optimized detection efficiency, contributing to the potential prevention of financial losses in real-world scenarios

Pneumonia Detection and Classification

Sept 2022

- Designed a Deep Learning Model using VGG16 with Transfer Learning to classify chest X-ray images as Pneumonia or Normal, achieving an accuracy of 91%.
- Leveraged Keras and TensorFlow to build and train a Convolutional Neural Network, enhancing medical imaging diagnostics through automated detection.

SOFT SKILLS

- Fast Learner
- Time Management
- Communication Skills
- Problem Solving